1.7 Lists

1.7.1 Collecting Objects Together

We first encountered lists in Section 1.2.3 as a way of collecting numbers together. You can use lists to collect together any kind of expression.

Here is a list of numbers.

```
In[1]:= {2, 3, 4}
Out[1]= {2, 3, 4}
```

This gives a list of symbolic expressions.

```
In[2]:= x^% - 1
                2        3        4
Out[2]= {-1 + x , -1 + x , -1 + x }
```

You can differentiate these expressions.

```
In[3]:= D[%, x]
                  2     3
Out[3]= {2 x, 3 x , 4 x }
```

Or you can find their values when `x` is replaced with 3.

```
In[4]:= % /. x -> 3
Out[4]= {6, 27, 108}
```

The mathematical functions that are built in to *Mathematica* are mostly set up so that they act separately on each element of a list. This is, however, not true of all functions in *Mathematica*. Unless you set it up specially, a new function `f` that you introduce will treat lists just as single objects. Sections 2.1.5 and 2.4.6 will describe how you can use `Map` and `Thread` to apply a function like this separately to each element in a list.